

Title: Deepfake Detection Using Convolutional Neural Networks: A Comparative Study

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Abstract:

This paper presents a deepfake detection system using transfer learning with MobileNetV2. We trained and evaluated two model versions on a subset of FaceForensics++ dataset consisting of 2000 frames. Our baseline model (V1) achieved 71.25% test accuracy, while the improved model (V2) achieved 74.50%. Robustness testing revealed sensitivity to both JPEG compression and Gaussian noise. A web-based demo was developed using Gradio for real-time inference.

1. Introduction

The proliferation of deepfake technology poses serious threats to digital media integrity. Detecting AI-generated facial manipulations has become a critical research area. In this work, we present a binary classification pipeline using MobileNetV2 pretrained on ImageNet, fine-tuned on FaceForensics++ data.

Our contributions are:

- A lightweight CNN-based deepfake detector
- Comparative analysis of two training configurations
- Robustness evaluation under image degradation
- A web-based inference interface
- **2. Related Work**
- Rossler et al. introduced FaceForensics++, a benchmark dataset for facial manipulation detection containing thousands of manipulated videos. Afchar et al. proposed MesoNet, a shallow CNN targeting mesoscopic image properties. Chollet's XceptionNet demonstrated state-of-the-art performance on deepfake detection by leveraging depthwise separable convolutions. Our work differs by prioritizing lightweight inference using MobileNetV2, making it suitable for resource-constrained environments.

- **3. Methodology**

- **3.1 Dataset**

- We used a subset of FaceForensics++ consisting of 400 videos — 200 real and 200 fake. Five frames were extracted per video at equal intervals, yielding 2000 images total (1000 real, 1000 fake). All frames were resized to 224×224 pixels.

- **3.2 Preprocessing**

- Two preprocessing pipelines were used. For V1, standard normalization was applied using ImageNet statistics. For V2, additional augmentation was applied during training including random horizontal flipping, random rotation up to 10 degrees, and color jitter.

Both pipelines normalize using mean=[0.485, 0.456, 0.406] and std=[0.229, 0.224, 0.225].

- **3.3 Model Architecture**
- MobileNetV2 pretrained on ImageNet was used as the backbone. The final classification layer was replaced with a binary linear layer outputting real/fake probabilities. The model contains approximately 3.4 million parameters.
- **3.4 Training Configuration**

Setting	V1	V2
Epochs	5	10
Learning Rate	0.001	0.0001
Optimizer	Adam	Adam
Loss	CrossEntropy	CrossEntropy
Batch Size	32	32

4. Experiments and Results

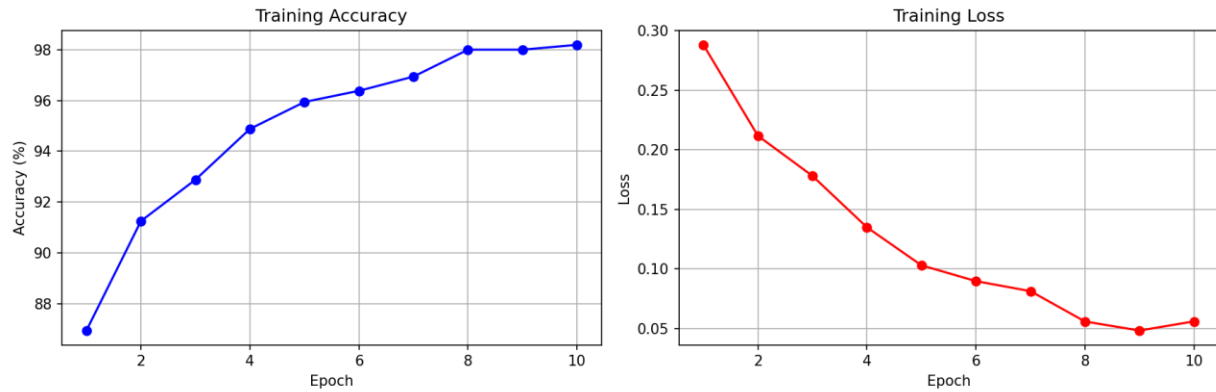
4.1 Training Performance

V1 training results:

Epoch	Loss	Train Accuracy
1	0.6302	63.44%
2	0.5163	73.44%
3	0.4955	76.19%
4	0.4619	77.56%
5	0.4191	79.75%

V2 training results:

Epoch	Loss	Train Accuracy
1	0.2881	86.94%
2	0.2117	91.25%
3	0.1780	92.88%
4	0.1348	94.88%
5	0.1029	95.94%
6	0.0896	96.38%
7	0.0812	96.94%
8	0.0557	98.00%
9	0.0481	98.00%
10	0.0558	98.19%



4.2 Test Performance

Model	Test Accuracy
V1 (Baseline)	71.25%
V2 (Improved)	74.50%

4.3 Classification Report (V2)

Class	Precision	Recall	F1-Score
Fake	0.71	0.78	0.74
Real	0.74	0.66	0.70
Accuracy			0.72

4.4 Robustness Testing

Condition	V1	V2
No degradation	71.25%	73.00%
JPEG Quality 50	70.50%	65.75%
JPEG Quality 20	65.75%	55.50%
Noise level 10	56.00%	61.00%
Noise level 25	58.25%	48.50%

5. Discussion

Several observations can be drawn from our experiments. First, reducing the learning rate from 0.001 to 0.0001 significantly improved training accuracy from 79.75% to 98.19%, though test accuracy improved modestly from 71.25% to 74.50%, indicating overfitting due to the limited dataset size.

Second, robustness testing revealed that V1 generalizes better under JPEG compression while V2 shows higher sensitivity, likely due to overfitting to clean training images. Gaussian noise remains a challenge for both models, suggesting that noise-based augmentation during training could improve robustness.

Third, the model shows limitations in detecting deepfakes outside the FaceForensics++ distribution, as demonstrated during real-world testing with the Gradio interface.

6. Conclusion and Future Work

This paper presented a deepfake detection system using MobileNetV2 transfer learning. Our best model achieved 74.50% test accuracy on FaceForensics++ subset. Robustness testing revealed sensitivity to image degradation, particularly Gaussian noise.

Future work includes:

- Training on larger and more diverse datasets
 - Applying noise augmentation to improve robustness
 - Experimenting with deeper architectures such as XceptionNet
 - Deploying the model as a permanent web service
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References

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